Lab 1 - Data Analysis and Modeling

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1 Introduction

This lab introduces the basics of R and RStudio. We will work with two built-in datasets, pressure and women, to practice data exploration, visualization, and analysis. We will complete 7 exercises and generate reproducible results in R Markdown format.

2 Exercise 1

Task: Complete the R language basics 1_3 from Cookbook for R _ Basics Status: [OK] Done

3 Exercise 2 - Exploring the pressure dataset

```
data(pressure)
head(pressure)
```

```
##   temperature pressure
## 1           0   0.0002
## 2          20   0.0012
## 3          40   0.0060
## 4          60   0.0300
## 5          80   0.0900
## 6         100   0.2700
```

3.1 Select specific data:

Fourth row, second column:

```
pressure[4, 2]
```

```
## [1] 0.03
```

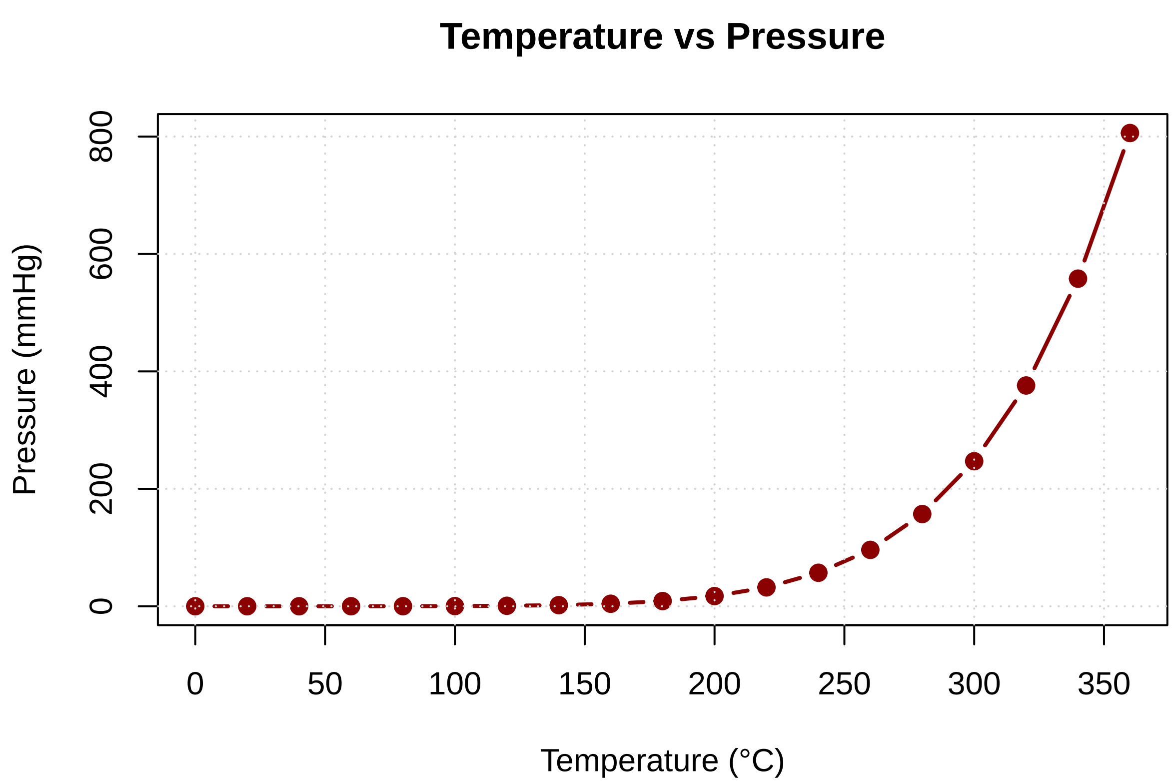
Select the entire second column:

```r
pressure[, 2]
```

```
##  [1]   0.0002   0.0012   0.0060   0.0300   0.0900   0.2700   0.7500   1.8500   4.2000   8.8000  17.3
## [17] 376.0000 558.0000 806.0000
```

4 Exercise 3 - Temperature vs. Pressure Plot

```r
plot(
  x = pressure$temperature,
  y = pressure$pressure,
  type = "b",
  col = "darkred",
  pch = 19,
  lwd = 2,
  xlab = "Temperature (°C)",
  ylab = "Pressure (mmHg)",
  main = "Temperature vs Pressure"
)
grid()
```



From the plot, we observe that as temperature increases, pressure also increases rapidly, showing a non-linear relationship.

5 Exercise 4 - Calculating Volume using PV = nRT

We use the ideal gas law:

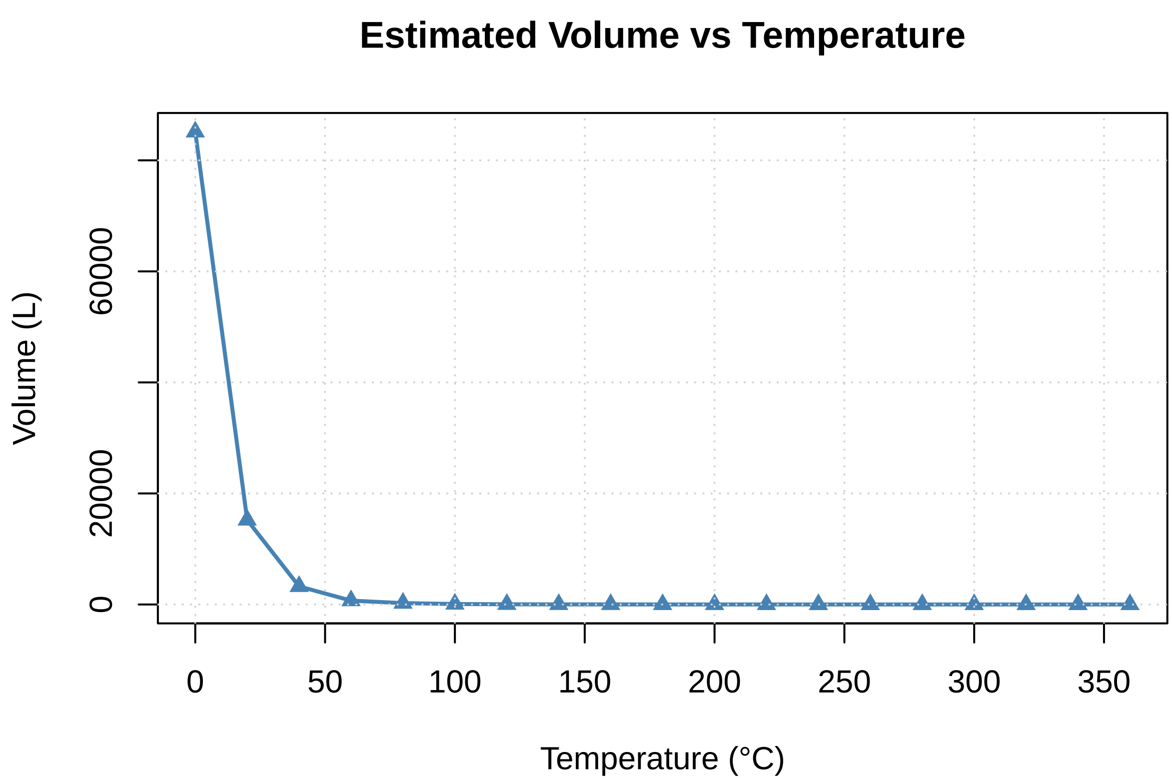
$$PV = nRT$$

Where: • (P): pressure in Pa • (T): temperature in Kelvin • (V): volume • (R = 8.31), (n = 1)

```
R <- 8.31
n <- 1
T <- pressure$temperature + 273.15
P <- pressure$pressure * 133.322  # mmHg to Pascal

V <- (n * R * T) / P
```

5.1 Plot: Temperature vs Volume

```
plot(
  x = pressure$temperature, y = V,
  type = "o", col = "steelblue", pch = 17, lwd = 2,
  xlab = "Temperature (°C)", ylab = "Volume (L)",
  main = "Estimated Volume vs Temperature"
)
grid()
```



This plot demonstrates the volume increases with temperature, as expected from gas law behavior.

6 Exercise 5 - Exploring the women dataset

```r
data(women)
head(women)
```

```
##   height weight
## 1     58    115
## 2     59    117
## 3     60    120
## 4     61    123
## 5     62    126
## 6     63    129
```

```r
dim(women)
```

```
## [1] 15  2
```

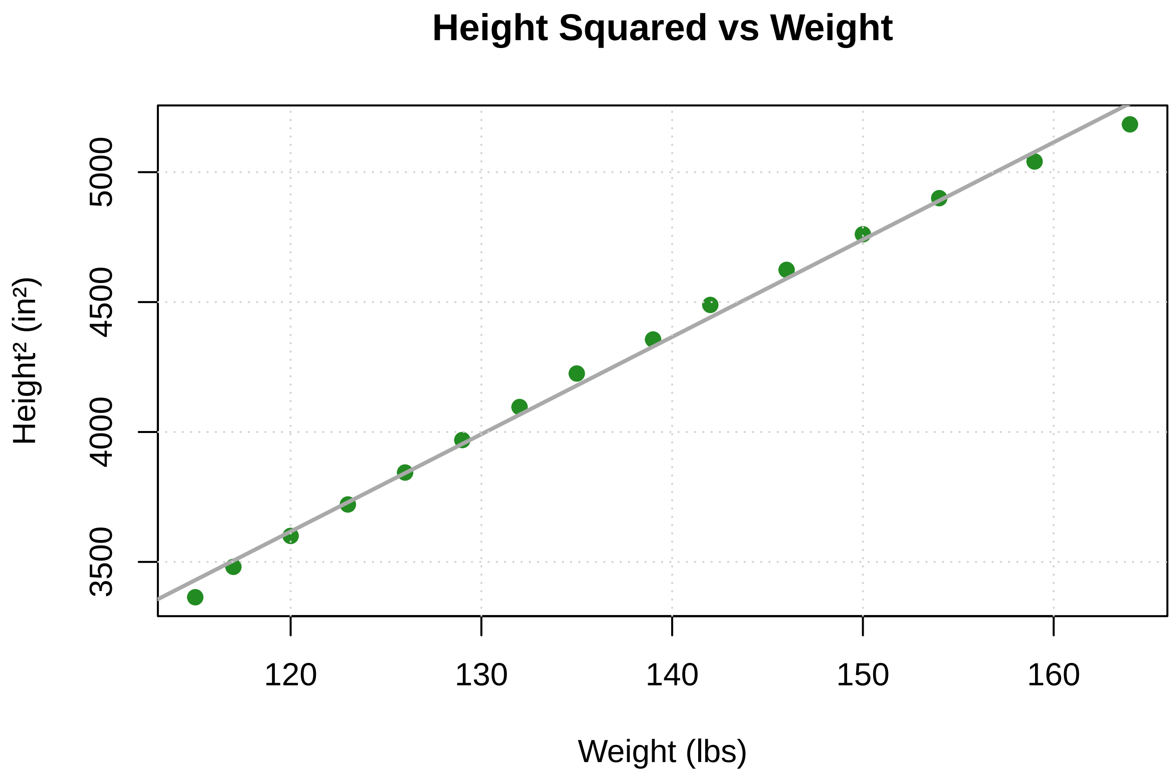
```r
names(women)
```

```
## [1] "height" "weight"
```

The dataset contains 15 rows and 2 columns: height (inches), weight (lbs)

7 Exercise 6 - Plot Height² vs Weight

```r
height_sq <- women$height^2

plot(
  x = women$weight, y = height_sq,
  type = "p", pch = 19, col = "forestgreen",
  xlab = "Weight (lbs)", ylab = "Height² (in²)",
  main = "Height Squared vs Weight"
)
abline(lm(height_sq ~ women$weight), col = "darkgray", lwd = 2)
grid()
```



We see a clear upward trend: as weight increases, squared height increases linearly.

8 Exercise 7 - BMI Calculation

Formula:

$$\text{BMI} = \frac{\text{weight (kg)}}{[\text{height (m)}]^2}$$

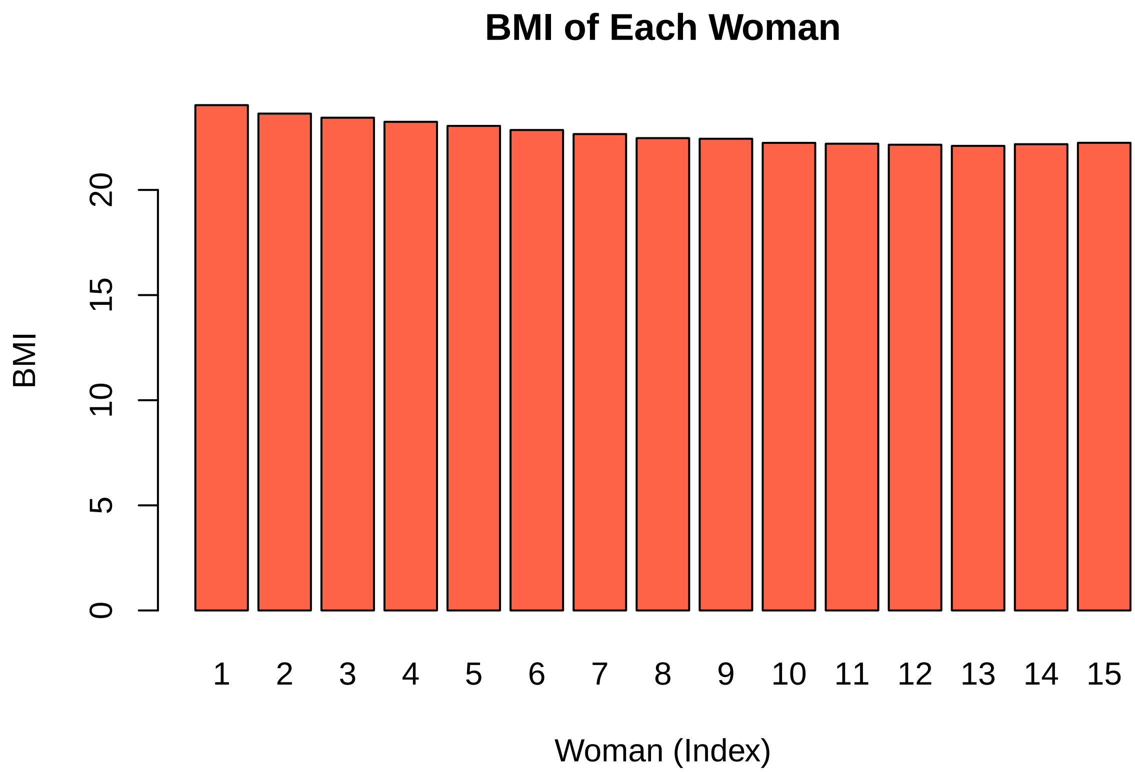
```
# Convert units
weight_kg <- women$weight * 0.453592
height_m <- women$height * 0.0254

BMI <- weight_kg / (height_m^2)
BMI
```

```
##  [1] 24.03476 23.63087 23.43563 23.24039 23.04545 22.85107 22.65750 22.46493 22.43494 22.24010 22.19
```

8.1 Plot BMI by individual

```
barplot(
  BMI,
  col = "tomato",
  names.arg = 1:15,
```

```r
  main = "BMI of Each Woman",
  xlab = "Woman (Index)", ylab = "BMI"
)
abline(h = 25, col = "blue", lty = 2)
text(x = which.max(BMI), y = max(BMI) + 1, labels = "Highest", col = "darkred")
```




```r
max_bmi_index <- which.max(BMI)
max_bmi_value <- max(BMI)
list("Index" = max_bmi_index, "Max BMI" = max_bmi_value)
```

```
## $Index
## [1] 1
##
## $`Max BMI`
## [1] 24.03476
```

Woman number r max_bmi_index has the highest BMI: r round(max_bmi_value, 2).

Reminder: Healthy BMI range is 18.5/text{-}24.9. This person may exceed that range.